

A Centralized Mobile Platform for CSE Department Clubs

1. Introduction

CSE Club Hub is a cross-platform mobile application designed specifically for the Computer Science and Engineering department. The app serves as a centralized digital platform for all five departmental clubs, enabling club executives to publish notices and announcements while allowing students to stay updated through a personalized, real-time feed.

Currently, club announcements are scattered across multiple WhatsApp groups, Facebook pages, and physical notice boards — leading to missed events, confusion, and poor student engagement. CSE Club Hub solves this by bringing all five clubs into one unified, role-based mobile application built with Flutter and powered by Firebase.

2. Problem Statement

The CSE department hosts five active student clubs, each running independently with no unified communication channel. This results in several recurring problems:

- Students miss important notices because they are spread across multiple platforms.
- There is no structured way to distinguish club executives from regular students.
- Events go unattended because reminders are inconsistent and unreliable.
- Club executives have no dedicated tool to manage their community posts.
- There is no centralized archive of past club activities and notices.

CSE Club Hub directly addresses all of these issues through a single, purpose-built application.

3. Project Objectives

1. Build a Flutter-based cross-platform mobile app for Android and iOS.
2. Integrate all five CSE department clubs into one unified platform.
3. Implement a role-based access system (Admin, Executive, Student).

4. Enable real-time push notifications for club announcements and events.
5. Provide executives with tools to post notices (text and image) and manage events.
6. Allow students to follow specific clubs and receive personalized updates.
7. Make the app deployable on the Google Play Store for real-world usage.

4. Project Scope

4.1 Clubs Covered

Index	Club Name	Focus Area
1	Machine Learning Club	AI, Data Science, Deep Learning
2	Competitive Programming Club	Algorithms, Contests, Problem Solving
3	IoT & Robotics Club	Hardware, Embedded Systems, Automation
4	Web Development Club	Frontend, Backend, Full Stack Web
5	Software Development Club	App Dev, System Design, Software Engineering
6	Cyber security Club	Cyber security

4.2 User Roles

Role	Permission
Super Admin	Manage all clubs, assign executives, remove any content
Club Executive	Post notices & events in own club, pin posts, manage club profile
Regular Students	Follow clubs, view posts, react & comment, RSVP to events

5. Core Features

5.1 Authentication & Profiles

- Email/password signup with student ID validation
- Profile setup: name, batch, section, department, avatar
- Role assignment managed by Super Admin

5.2 Club Feed

- Personalized home feed showing posts from followed clubs
- Department-wide feed showing all clubs' posts
- Post creation with text and image upload (executives only)
- Pin important posts to the top of the feed
- Reactions (Like, Fire, Clap) and comment threads

5.3 Events Module

- Executives can create events with title, description, date, time, and venue
- Students can RSVP as Going or Interested
- Automatic push notification reminder 24 hours before the event

5.4 Smart Notifications

- Instant push notification when a followed club posts a notice
- Event reminders sent automatically via Firebase Cloud Messaging
- Notification center with read/unread status

5.5 Search & Discovery

- Search posts, events, clubs, and users
- Browse all clubs and follow/unfollow with one tap

6. Technical Architecture

6.1 Technology Stack

Layer	Technology	Purpose
Frontend	Flutter (Dart)	Cross-platform UI for Android & iOS
Backend	Firebase Firestore	Real-time NoSQL cloud database
Authentication	Firebase Auth	Secure login and role management
File Storage	Firebase Storage	Image uploads for posts and profiles
Notifications	Firebase Cloud Messaging	Push notifications for all users

7. Expected Outcomes

- A fully functional, publishable Flutter mobile application for Android.
- All five CSE clubs are onboarded with real executive accounts pre-configured.
- Real-time notices and event announcements are replacing fragmented WhatsApp groups.
- Demonstrated use of Flutter, Firebase, real-time streams, and role-based access control.
- A deployable app on the Google Play Store for genuine university community use.

10. Conclusion

CSE Club Hub represents a practical, impactful, and technically rich mobile application project that aligns with the goals of the Mobile Application Development course. With a clear problem statement, defined scope, realistic timeline, and real-world deployment ambition, this project stands as both a strong academic submission and a genuine contribution to the CSE department community.